

Long–Short Portfolio Return Book

Cumulative return of each factor's long–short portfolio, net of 20 bp transaction cost.
227 factors over 200 months, ordered by in-sample L–S CAGR (last 48 months, net of cost).

HOW TO READ

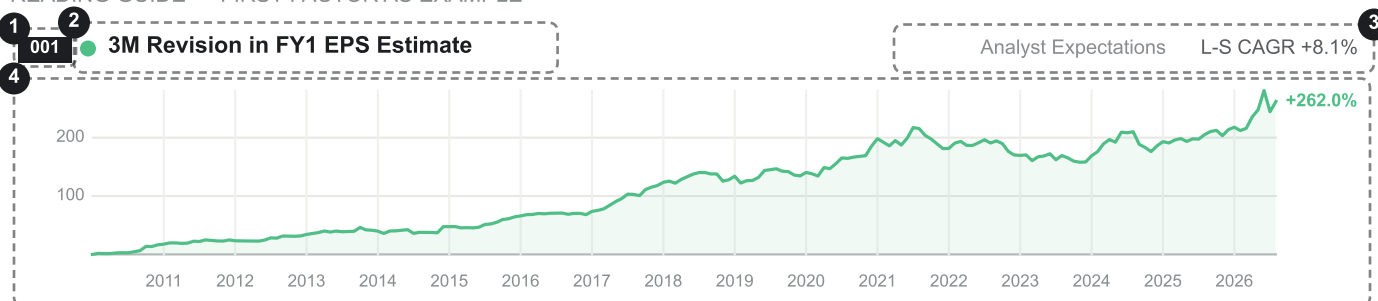
Each curve starts at 0 % in January 2010 and compounds monthly long–short returns. The line is coloured by the factor's style; the figure at the right edge is the period-end cumulative return.

FACTOR STYLES

Valuation	35	Price Momentum	50
Earnings Quality	47	Size	4
Analyst Expectations	18	Historical Growth	38
Capital Efficiency	27	Volatility	8

001 In current model portfolio (29 factors)

READING GUIDE — FIRST FACTOR AS EXAMPLE



1 Rank - ordered by ③ L-S CAGR, descending.
Filled dark badge = in the current Top-50 portfolio.

2 Factor name - colour dot = style (legend above).

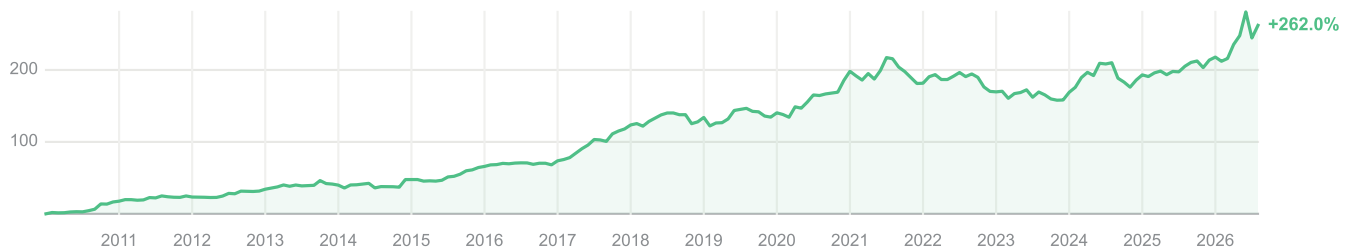
3 Style name / L-S CAGR - the LAST 48 MONTHS only (rolling IS), net of 20 bp cost. The live selection / sort basis. The curve below uses these same live 48M L/S rules.

4 Curve - the live 48M L/S rules applied over FULL HISTORY (Jul 2015 - Aug 2026), net of 20 bp cost. Rules are fit on recent data, so the long-run curve is context, not a live track record - it can rise while the recent ③ is negative.

001 ● 3M Revision in FY1 EPS Estimate

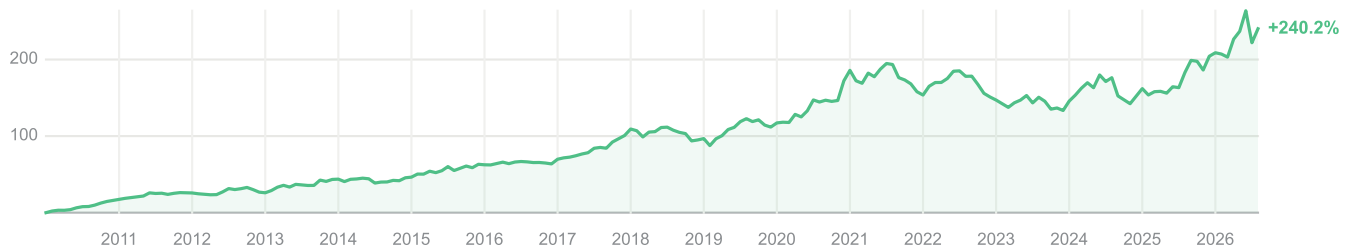
Analyst Expectations

CAGR +8.1%

**002 ● 3M Revision in FY2 EPS Estimate**

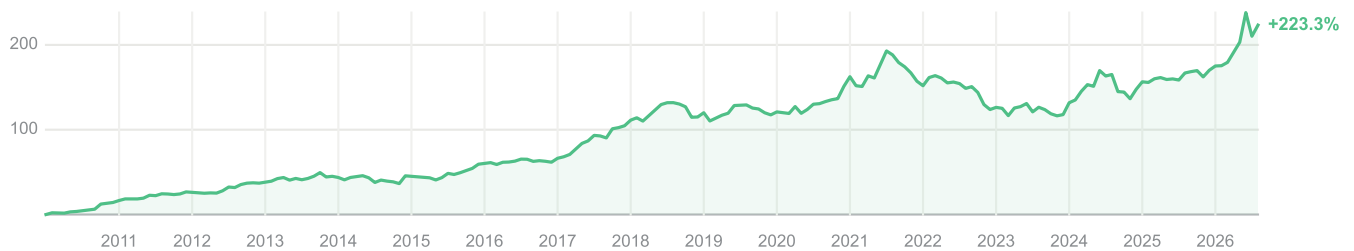
Analyst Expectations

CAGR +7.7%

**003 ● Street Revision Magnitude**

Analyst Expectations

CAGR +7.3%

**004 ● Number of EPS FY1 Revisions**

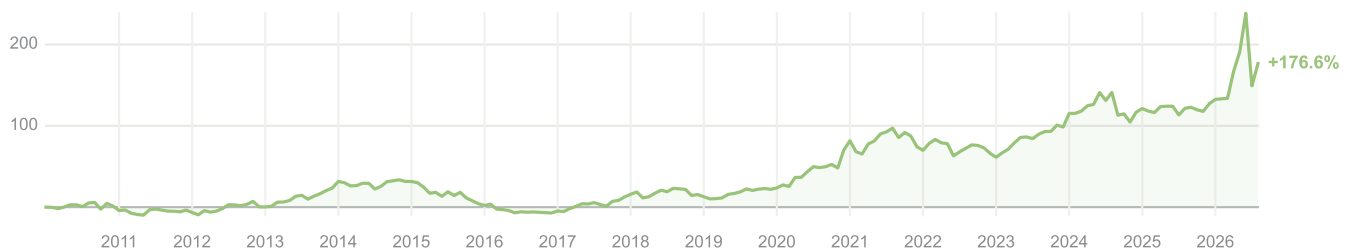
Analyst Expectations

CAGR +6.8%

**005 ● Slope of 52W Price Trend**

Price Momentum

CAGR +6.3%

**006 ● Adjusted Revision Magnitude**

Analyst Expectations

CAGR +6.3%



007 ● **6M Chg in Target Price**

Analyst Expectations

CAGR +6.3%

008 ● **Standardized Unexpected Earnings**

Analyst Expectations

CAGR +5.8%

009 ● **Closing Price to 260 Day Low**

Price Momentum

CAGR +5.6%

010 ● **Cash Flow Return on Invested Capital**

Capital Efficiency

CAGR +5.2%

011 ● **3Y Average Annual Earnings Growth**

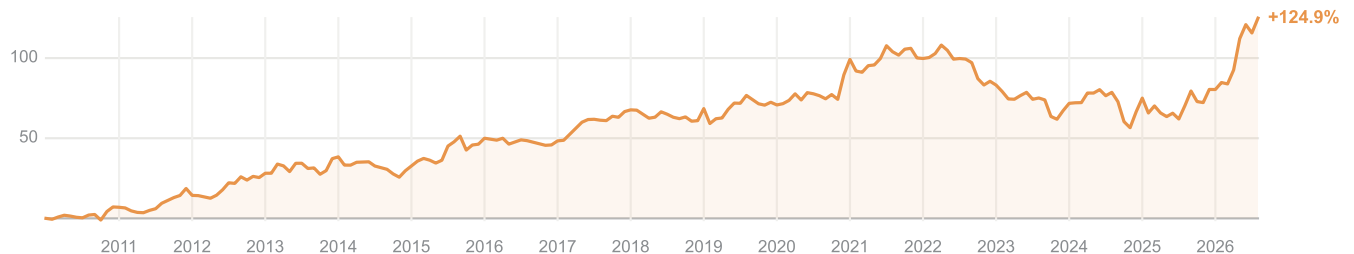
Historical Growth

CAGR +5.1%

012 ● **3Y Chg in Earnings per Share**

Historical Growth

CAGR +5.0%



013 ● Sharpe Ratio

Price Momentum

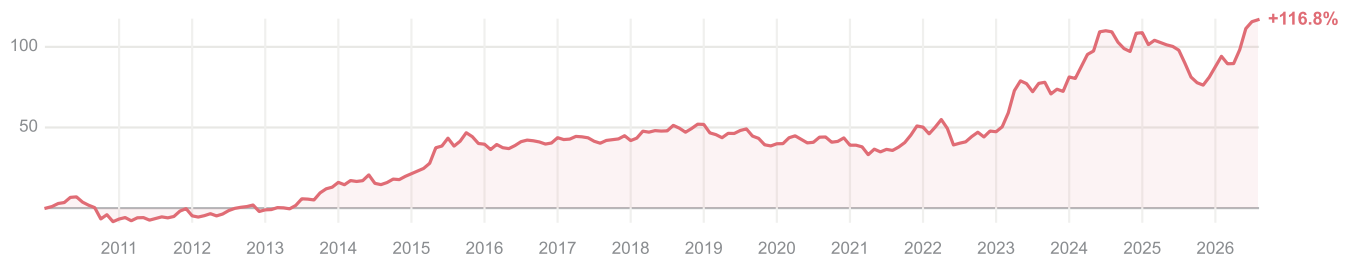
CAGR +5.0%



014 ● ROE 20 Qtr Standard Deviation

Earnings Quality

CAGR +4.8%



015 ● Operating Cash Flow to Enterprise Value

Valuation

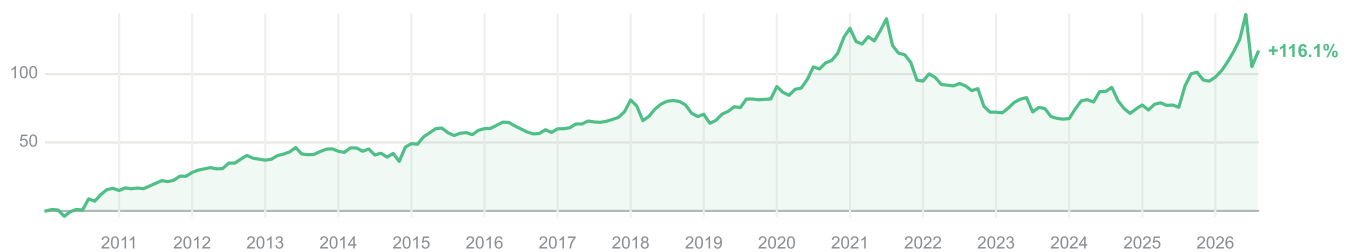
CAGR +4.8%



016 ● Adjusted Number of EPS FY1 Revisions

Analyst Expectations

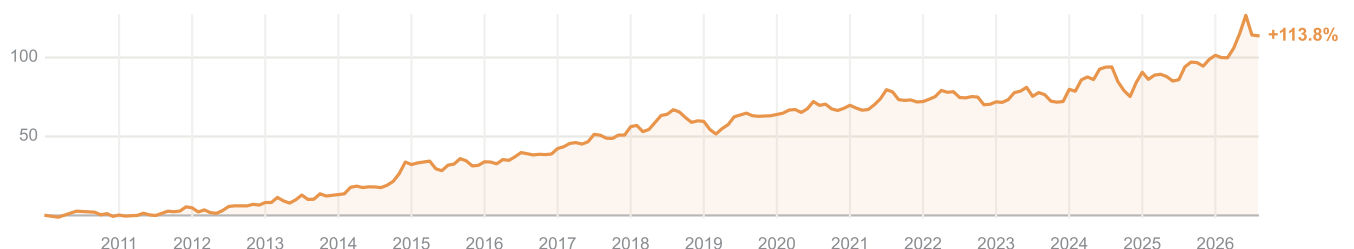
CAGR +4.8%



017 ● 1Y Chg in Sales Turnover

Historical Growth

CAGR +4.7%



018 ● 24M Residual Return Variance

Price Momentum

CAGR +4.5%



019 ● ROA 20 Qtr Standard Deviation

Earnings Quality

CAGR +4.4%

**020 ● Net Profit Margin**

Earnings Quality

CAGR +4.4%

**021 ● Operating Cash Flow to Equity**

Capital Efficiency

CAGR +4.4%

**022 ● 1Y Chg in Operating Cash Flow per Share**

Historical Growth

CAGR +4.2%

**023 ● Operating Cash Flow Ratio**

Capital Efficiency

CAGR +4.2%

**024 ● Free Cash Flow to Sales**

Earnings Quality

CAGR +4.2%



025 ● Long Term Debt to Equity Ratio

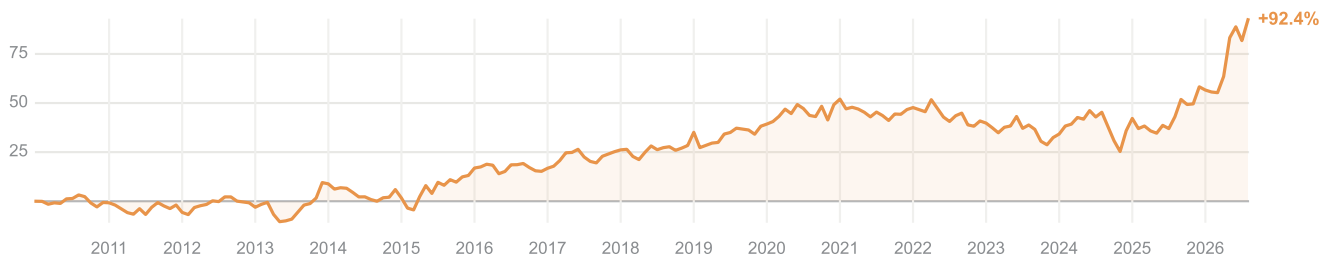
Capital Efficiency

CAGR +4.2%

**026 ● 3Y Chg in Operating Margin**

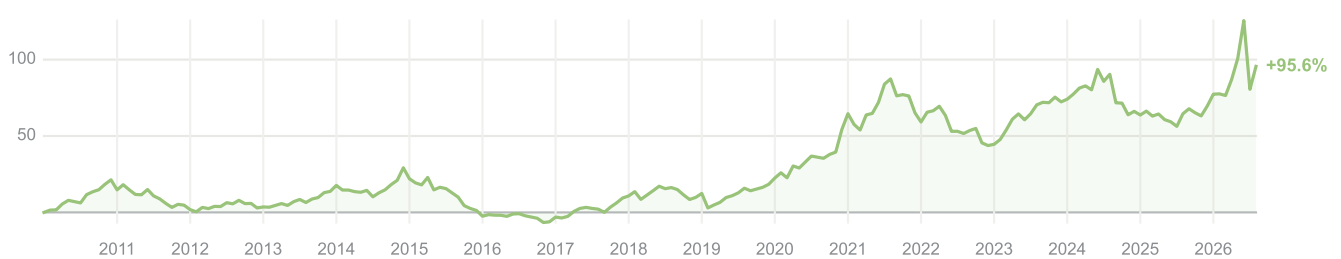
Historical Growth

CAGR +4.1%

**027 ● 180D Price TStat**

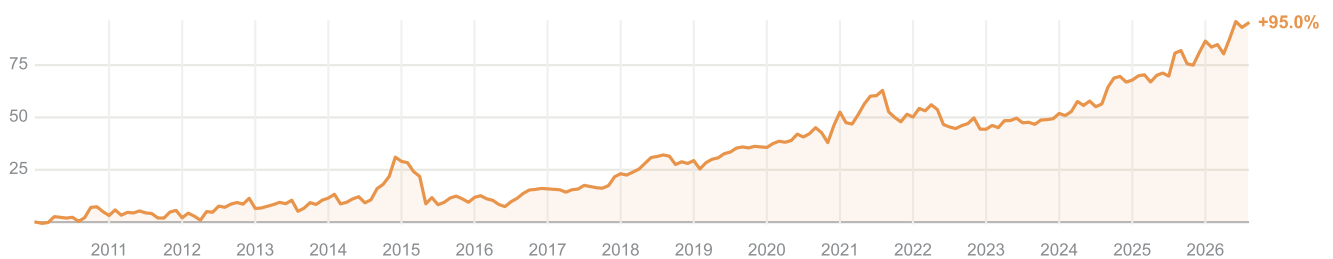
Price Momentum

CAGR +4.1%

**028 ● 1Y Chg in Asset Adjusted Free Cash Flow**

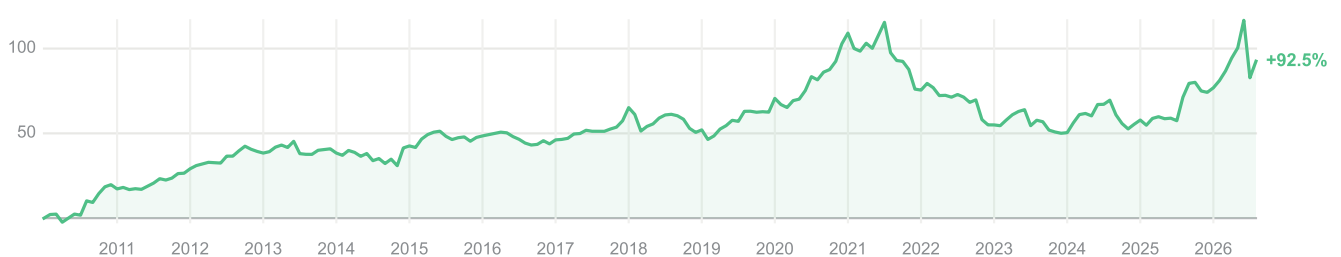
Historical Growth

CAGR +4.1%

**029 ● Analyst Earnings Estimate Diffusion**

Analyst Expectations

CAGR +4.0%

**030 ● Sales to Price Ratio**

Valuation

CAGR +4.0%



031 ● Free Cash Flow to Common Equity

Valuation

CAGR +3.9%



032 ● EBIT Margin

Earnings Quality

CAGR +3.8%



033 ● Gross Profit Margin

Earnings Quality

CAGR +3.8%



034 ● Sales to EV Ratio

Valuation

CAGR +3.8%



035 ● Asset Turnover

Earnings Quality

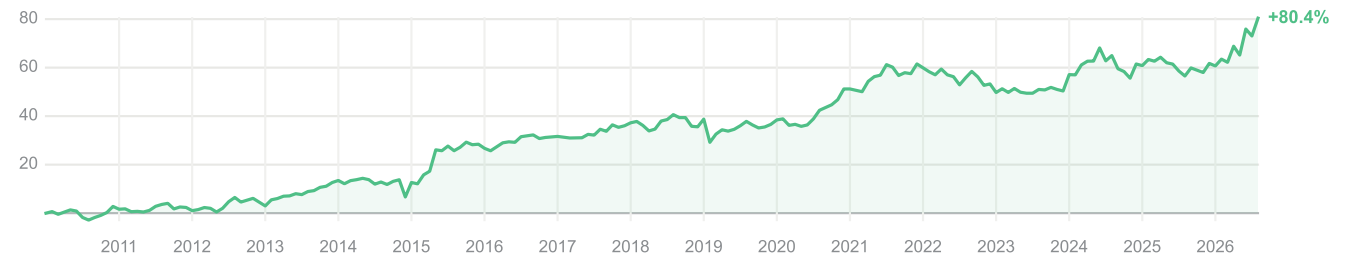
CAGR +3.7%



036 ● 6M Avg Chg 1M Recommendation

Analyst Expectations

CAGR +3.6%



037 ● **Institution Ownership Concentration**

Price Momentum

CAGR +3.6%

038 ● **Target Price Gap to 6M EMA**

Analyst Expectations

CAGR +3.6%

039 ● **Current Liability to Price**

Valuation

CAGR +3.6%

040 ● **Operating Cash Flow to Price**

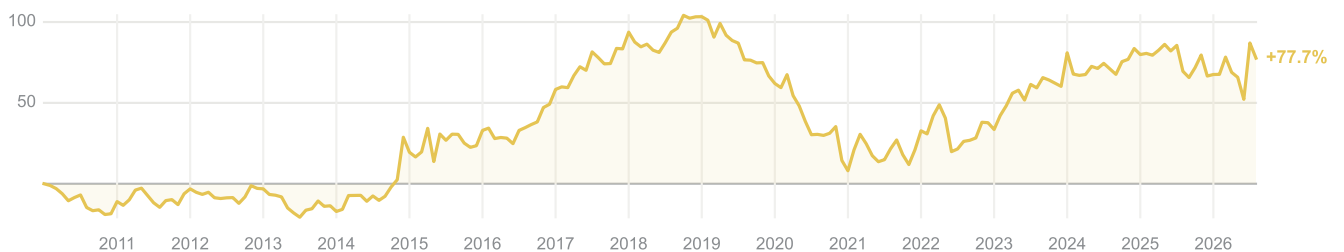
Valuation

CAGR +3.6%

041 ● **Tobin's Q Ratio**

Capital Efficiency

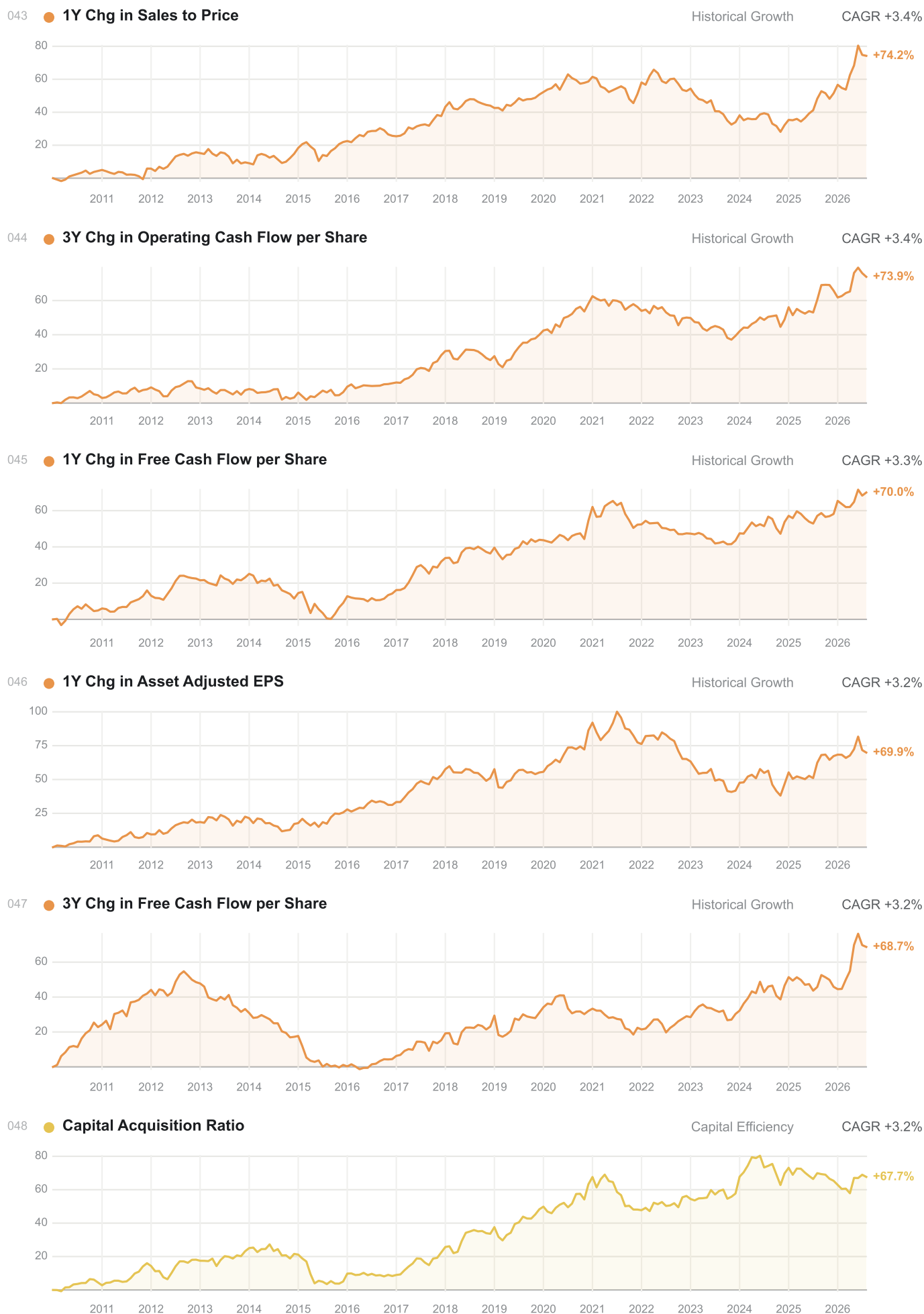
CAGR +3.5%

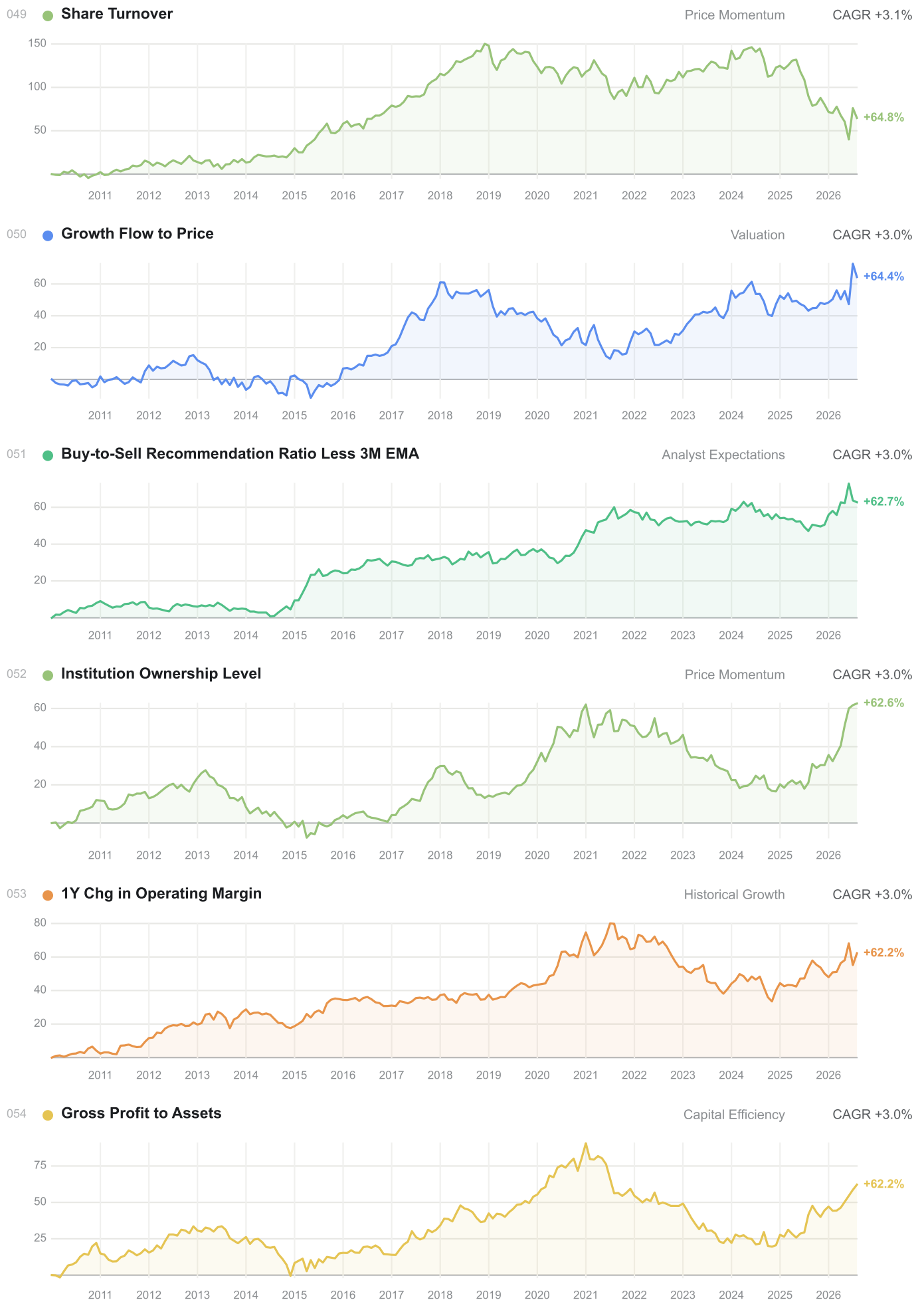
042 ● **Adjusted Accruals**

Earnings Quality

CAGR +3.5%







055 ● 1Y Chg in Cash Flow per Share

Historical Growth

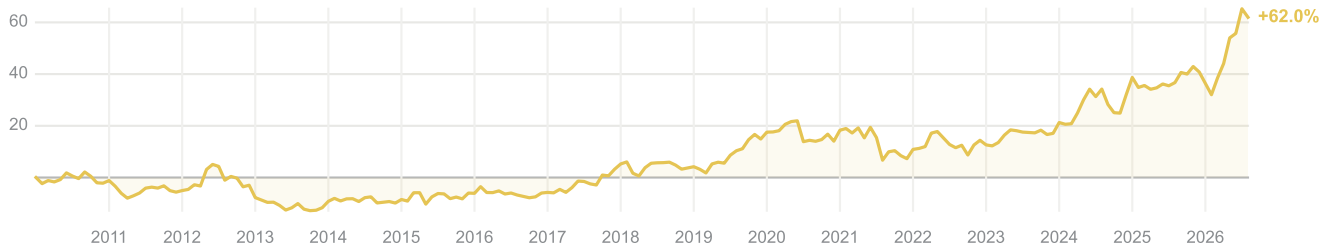
CAGR +3.0%



056 ● Receivables Turnover Ratio

Capital Efficiency

CAGR +3.0%



057 ● 1Y Chg in Asset Adjusted Cash Flow

Historical Growth

CAGR +2.9%



058 ● 3Y Chg in Sales Turnover

Historical Growth

CAGR +2.9%



059 ● 1Y Chg in Cash Flow to Price

Historical Growth

CAGR +2.9%



060 ● 6 Months Price Reversal

Price Momentum

CAGR +2.9%



061 ● Equity Turnover

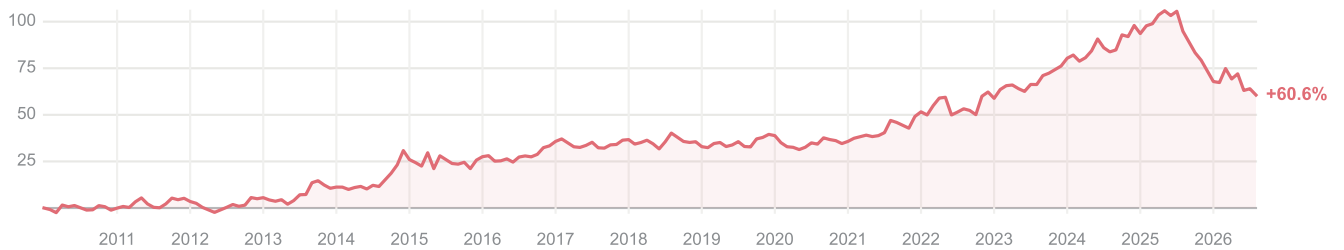
Earnings Quality

CAGR +2.9%

**062 ● Working Capital Accruals**

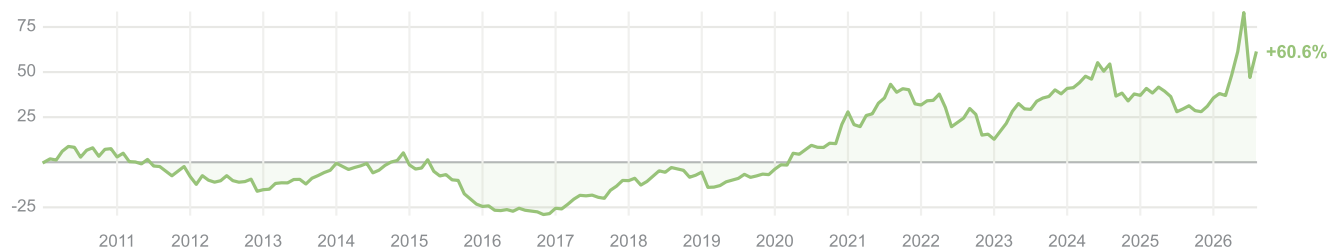
Earnings Quality

CAGR +2.9%

**063 ● 15/36 Week Price Ratio**

Price Momentum

CAGR +2.9%

**064 ● Operating Margin**

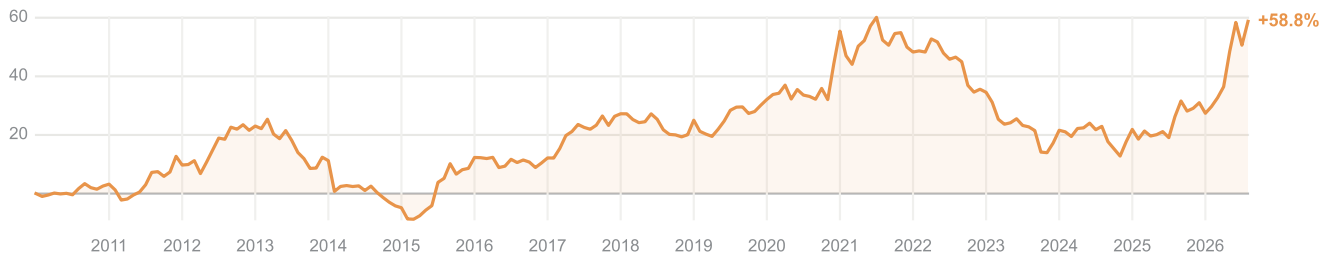
Earnings Quality

CAGR +2.9%

**065 ● 3Y Chg in Asset Adjusted Cash Flow**

Historical Growth

CAGR +2.8%

**066 ● 1Y Chg in Asset Adjusted Operating Cash Flow**

Historical Growth

CAGR +2.8%



067 ● Risk Adjusted Relative Strength

Price Momentum

CAGR +2.8%



068 ● Institution Ownership Turnover

Price Momentum

CAGR +2.8%



069 ● Capital Expenditure to Total Assets

Capital Efficiency

CAGR +2.7%



070 ● 6M Chg in Target Price Gap

Analyst Expectations

CAGR +2.7%



071 ● Assets to Price Ratio

Valuation

CAGR +2.7%



072 ● Unexpected Receivables Change

Earnings Quality

CAGR +2.7%



073 ● 1Y Chg in Earnings to Price

Historical Growth

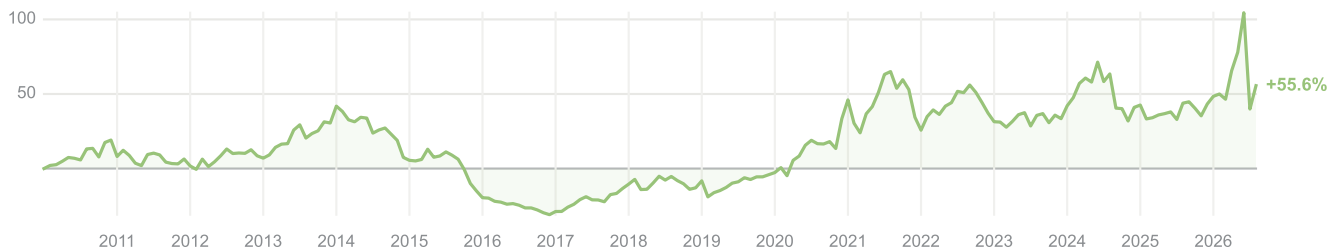
CAGR +2.7%



074 ● 12M - 1M Price Momentum

Price Momentum

CAGR +2.7%



075 ● Sales to Invested Capital

Earnings Quality

CAGR +2.6%



076 ● Change EPS to Sales

Earnings Quality

CAGR +2.6%



077 ● 1Y Chg in Earnings per Share

Historical Growth

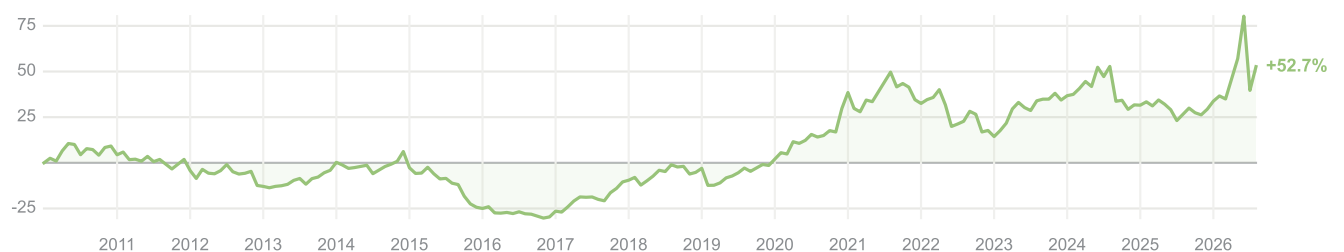
CAGR +2.6%



078 ● 50 Day to 200 Day Stock Price

Price Momentum

CAGR +2.6%



079 ● **Ownership Breadth Stability of Institution Ownership by Holders**

Price Momentum

CAGR +2.6%

080 ● **Free Cash Flow to Price**

Valuation

CAGR +2.5%

081 ● **Book to Price**

Valuation

CAGR +2.5%

082 ● **3Y Chg in Asset Adjusted Operating Cash Flow**

Historical Growth

CAGR +2.5%

083 ● **Net Cash Flow to Price**

Valuation

CAGR +2.5%

084 ● **Cash Flow to Equity**

Valuation

CAGR +2.5%



085 ● Long Term Debt to Assets Ratio

Capital Efficiency

CAGR +2.4%



086 ● Working Capital to Assets Ratio

Earnings Quality

CAGR +2.4%



087 ● 3Y Chg in Operating Cash Flow to Price

Historical Growth

CAGR +2.4%



088 ● Dividends to Price Ratio

Valuation

CAGR +2.3%



089 ● Cash to Price

Valuation

CAGR +2.3%



090 ● Accrual Ratio - Balance Sheet

Earnings Quality

CAGR +2.3%



091 ● **Cash to EV Ratio**

Valuation

CAGR +2.3%

092 ● **Adjusted EBIT to Price**

Valuation

CAGR +2.3%

093 ● **Depreciation to Capex Ratio**

Earnings Quality

CAGR +2.2%

094 ● **Pretax Income to Net Operating Assets**

Earnings Quality

CAGR +2.2%

095 ● **Free Cash Flow to Enterprise Value**

Valuation

CAGR +2.2%

096 ● **Cash to Total Assets**

Valuation

CAGR +2.2%



097 ● **Log of Unadjusted Stock Price**

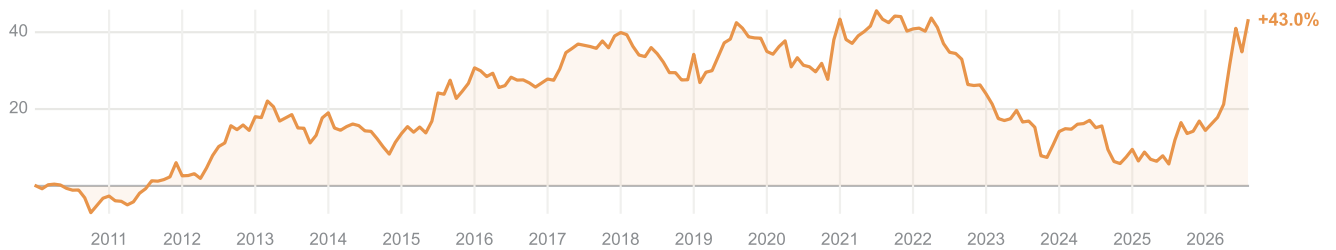
Price Momentum

CAGR +2.2%

098 ● **3Y Chg in Asset Adjusted EPS**

Historical Growth

CAGR +2.2%

099 ● **Forward Earnings to Price**

Valuation

CAGR +2.2%

100 ● **Cash Flow to Total Asset**

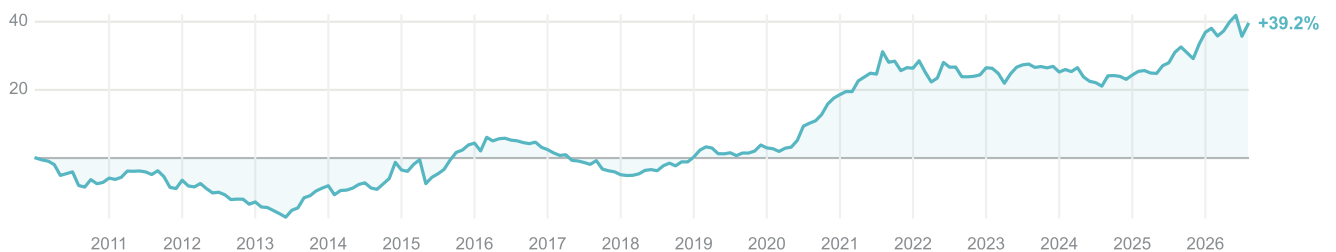
Earnings Quality

CAGR +2.2%

101 ● **60M CAPM Beta**

Volatility

CAGR +2.2%

102 ● **Retained Earnings to Total Assets**

Earnings Quality

CAGR +2.1%



103 ● Pretax Profit Margin

Capital Efficiency

CAGR +2.1%



104 ● 26W Price Stat

Price Momentum

CAGR +2.1%



105 ● Solvency Ratio

Valuation

CAGR +2.1%



106 ● Accrual Ratio - Cash Flows

Earnings Quality

CAGR +2.1%



107 ● Interest Coverage Ratio

Capital Efficiency

CAGR +2.1%



108 ● Net Cash Flow to Enterprise Value

Valuation

CAGR +2.1%



109 ● Earnings to Price

Valuation

CAGR +2.1%



110 ● Adjusted Forward Earnings to Price

Valuation

CAGR +2.0%



111 ● Dividend Growth

Historical Growth

CAGR +2.0%



112 ● Return on Assets

Capital Efficiency

CAGR +2.0%



113 ● Operating Earnings to Assets Ratio

Earnings Quality

CAGR +2.0%

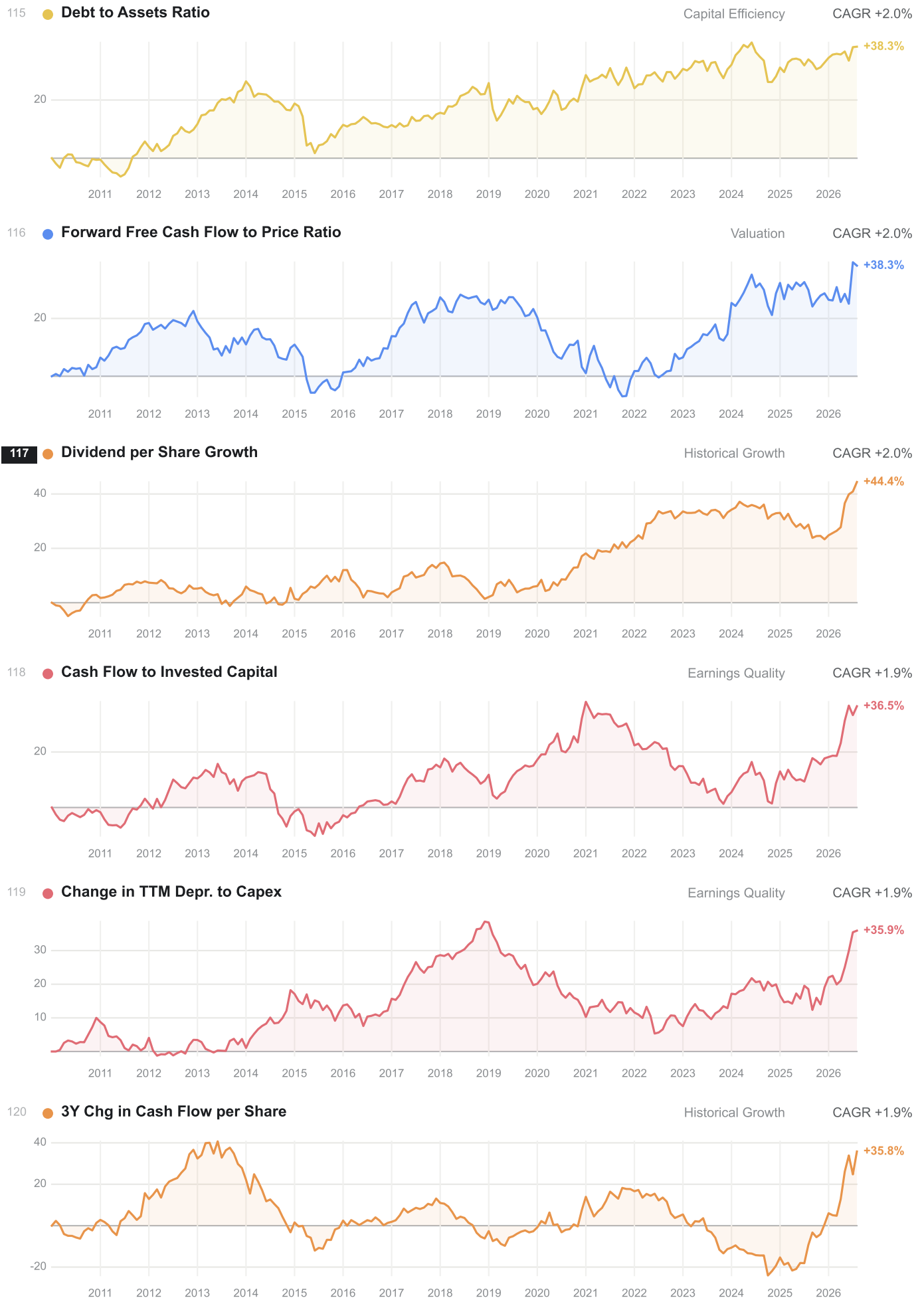


114 ● Unexpected Inventory Change

Earnings Quality

CAGR +2.0%

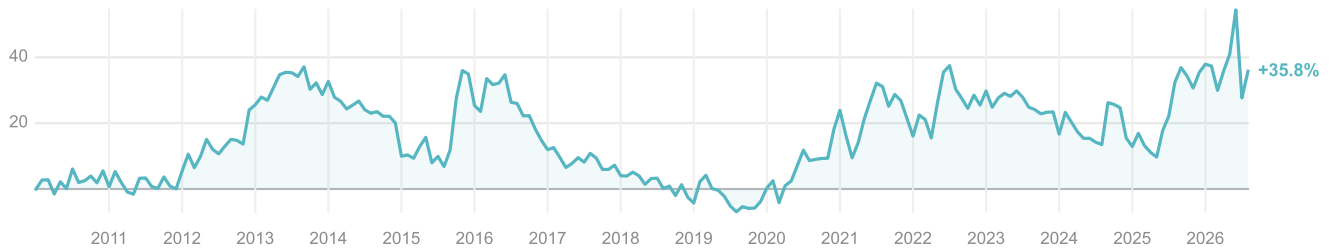




121 ● 90 Day Coefficient of Variation

Volatility

CAGR +1.9%



122 ● Inverse P/E Ratio Adj for Growth and Yield

Valuation

CAGR +1.8%



123 ● EBITDA to Enterprise Value

Valuation

CAGR +1.8%



124 ● Return on Equity

Capital Efficiency

CAGR +1.8%



125 ● 60M CAPM Alpha

Price Momentum

CAGR +1.8%



126 ● 50 Day Volume Signal

Price Momentum

CAGR +1.8%



127 ● **EBITDA to Assets**

Capital Efficiency

CAGR +1.8%

128 ● **Net Current Assets to Price Ratio**

Valuation

CAGR +1.8%

129 ● **Year over Year Change of Inventory to Assets**

Earnings Quality

CAGR +1.8%

130 ● **Year over Year Change of EPS to Sales**

Earnings Quality

CAGR +1.8%

131 ● **Operating Earnings to Price Ratio**

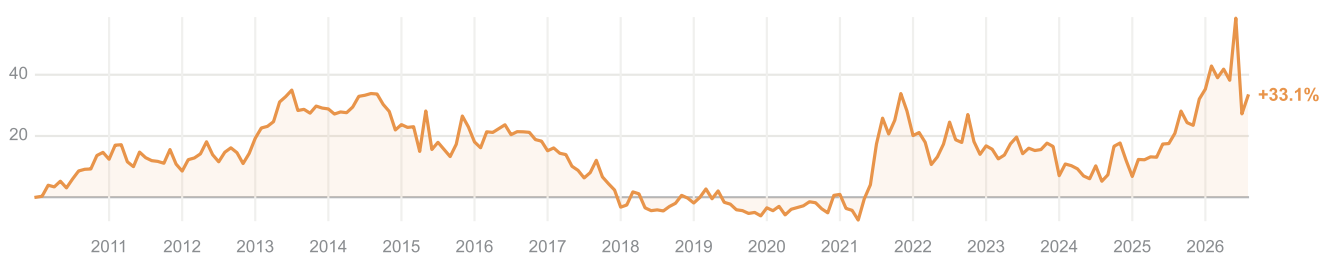
Valuation

CAGR +1.8%

132 ● **Trailing 12-Month Receivables and Inventories/Trailing 12-Month Sales**

Historical Growth

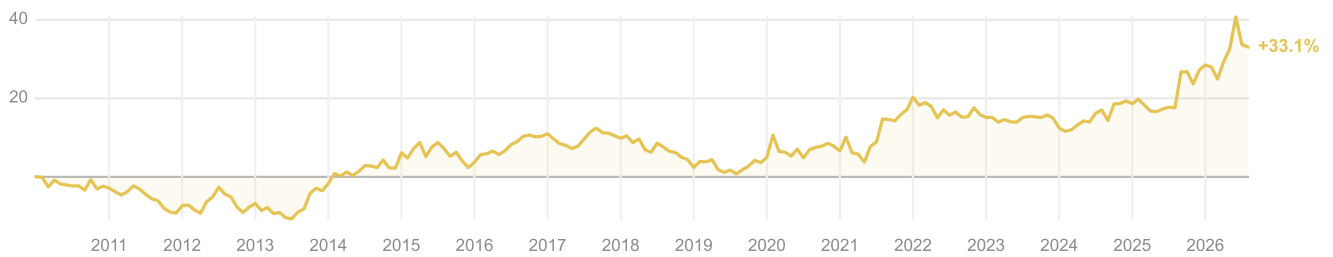
CAGR +1.7%



133 ● Adjusted Interest Coverage Ratio

Capital Efficiency

CAGR +1.7%

**134 ● Capital Expenditure to Sales**

Earnings Quality

CAGR +1.7%

**135 ● Current Ratio**

Earnings Quality

CAGR +1.7%

**136 ● Normalized EP**

Valuation

CAGR +1.7%

**137 ● Max Daily Return in the Past 6 Months**

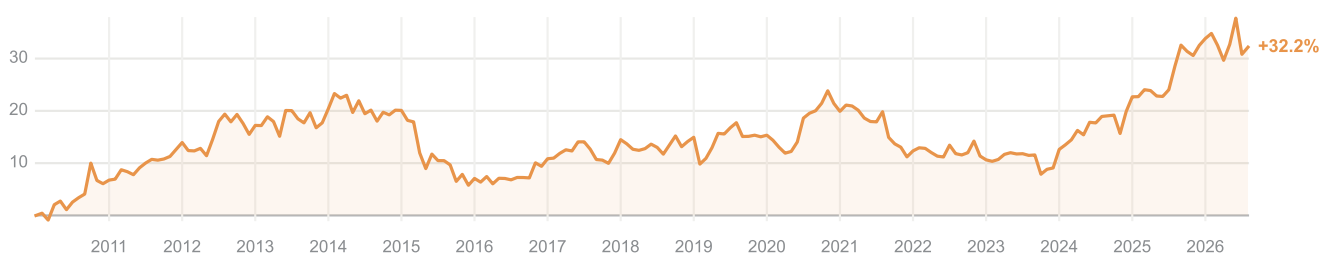
Price Momentum

CAGR +1.7%

**138 ● 3Y Chg in Free Cash Flow to Price**

Historical Growth

CAGR +1.7%



139 ● **Institution Ownership Level - Active Manager**

Price Momentum

CAGR +1.7%

140 ● **Foreign Institution Ownership**

Price Momentum

CAGR +1.7%

141 ● **Year over Year Change in SGA to Sales**

Earnings Quality

CAGR +1.7%

142 ● **SG&A to Sales**

Earnings Quality

CAGR +1.7%

143 ● **Book Leverage**

Volatility

CAGR +1.6%

144 ● **Pretax Income Per Share to Price**

Valuation

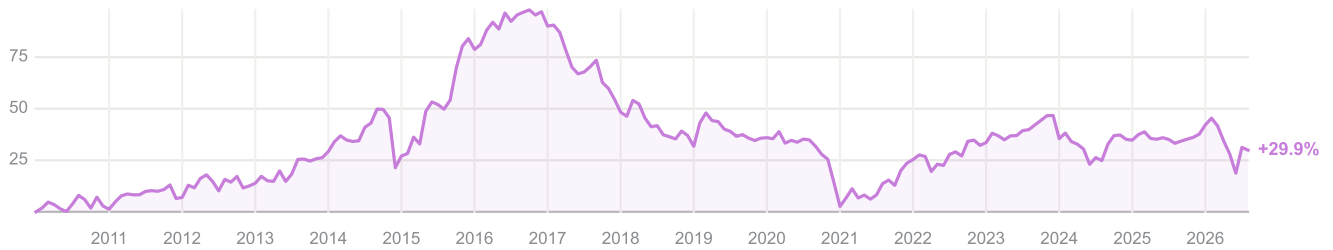
CAGR +1.6%



145 ● **Log of Market Capitalization Cubed**

Size

CAGR +1.6%

146 ● **3Y Chg in Asset Adjusted Free Cash Flow**

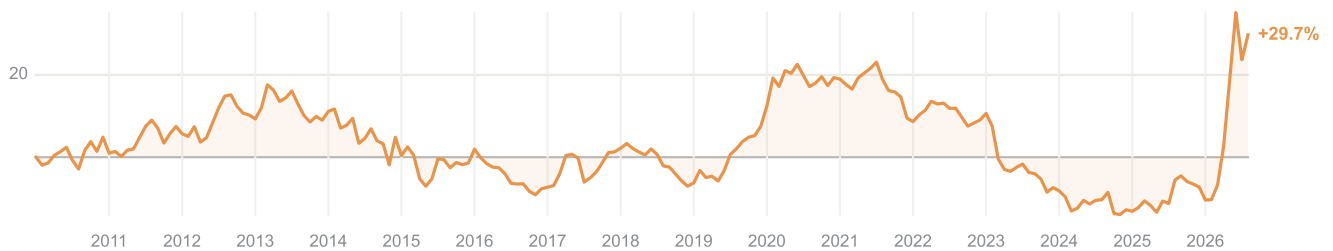
Historical Growth

CAGR +1.6%

147 ● **3Y Average Annual Sales Growth**

Historical Growth

CAGR +1.6%

148 ● **Buy-to-Sell Recommendation Ratio Less 3M SMA**

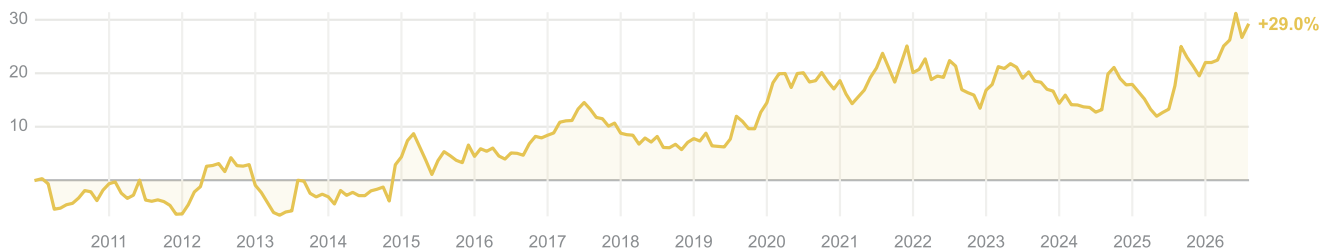
Analyst Expectations

CAGR +1.6%

149 ● **Working Capital Turnover Ratio**

Capital Efficiency

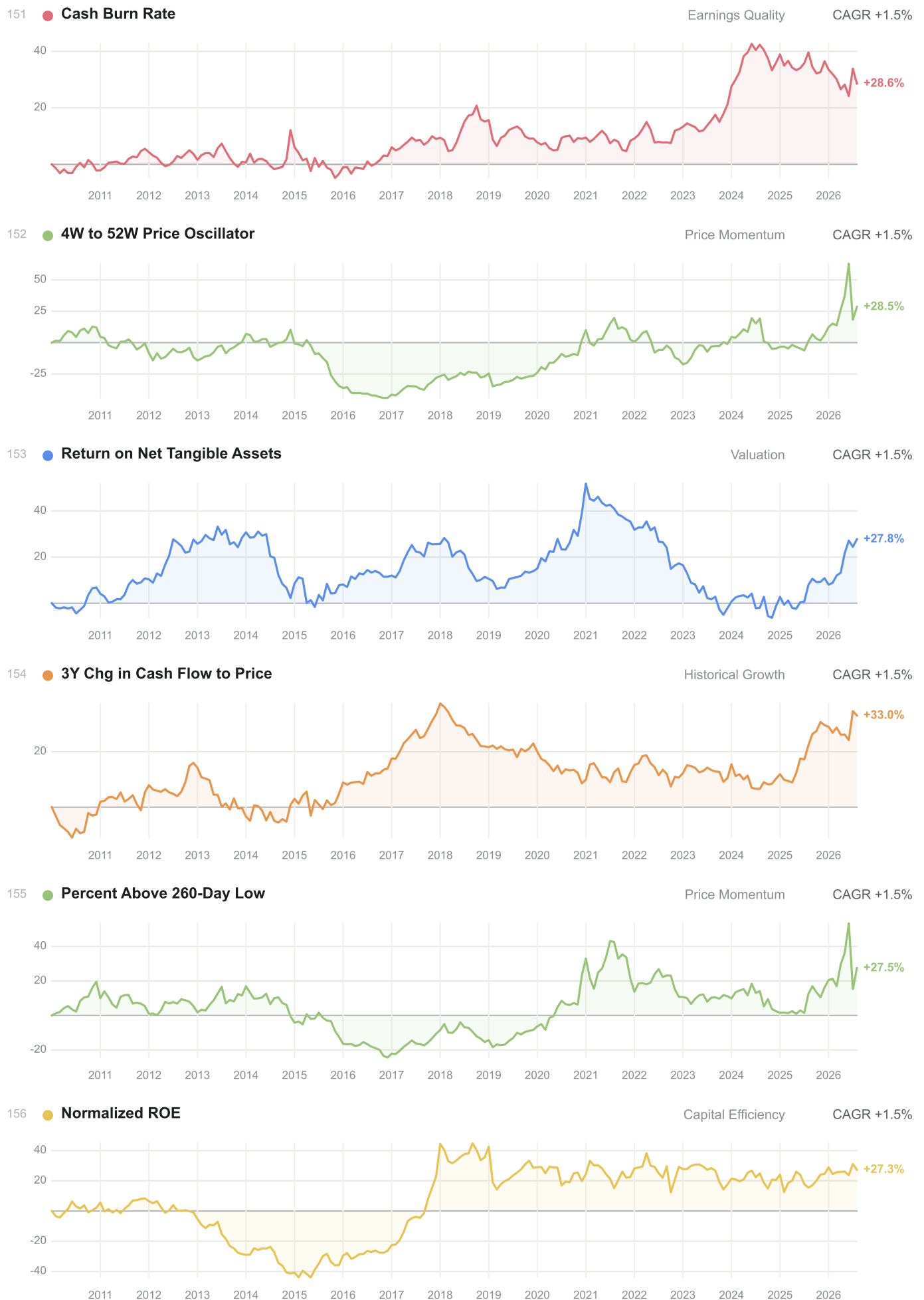
CAGR +1.5%

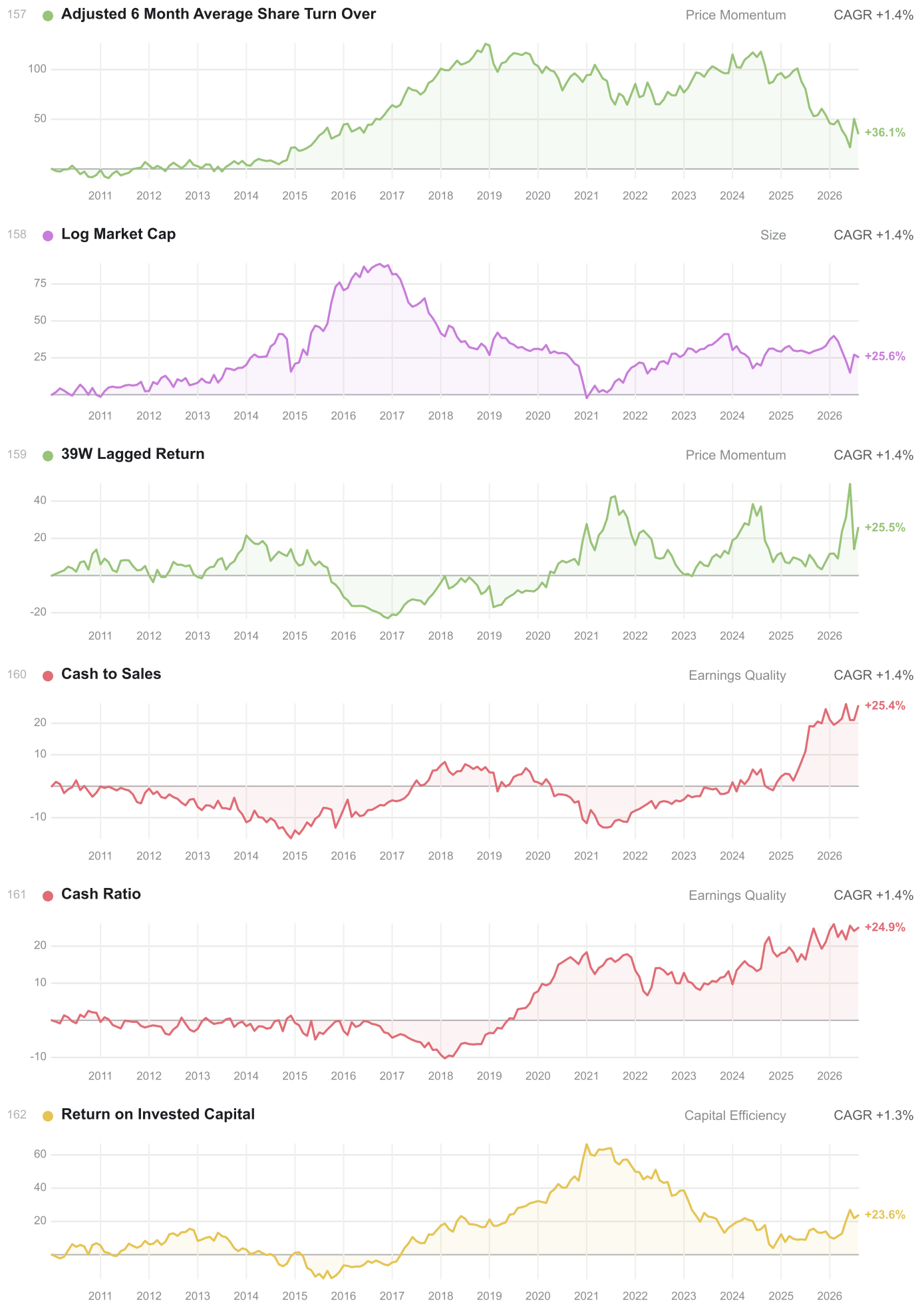
150 ● **Working Capital to Total Assets**

Earnings Quality

CAGR +1.5%







163 ● 6 Month Average Share Turn Over

Price Momentum

CAGR +1.3%

**164 ● 8-Week Change in 12-Month Forward Earnings Consensus Estimate/Price**

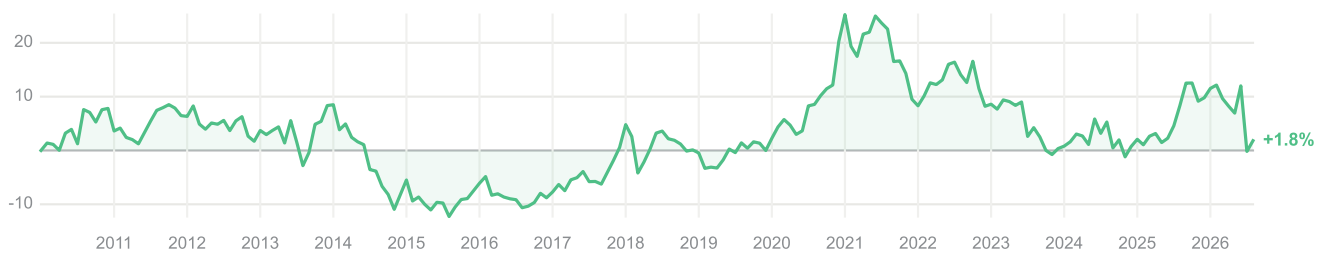
Valuation

CAGR +1.2%

**165 ● Adj Number of EPS FY2 Revisions**

Analyst Expectations

CAGR +1.2%

**166 ● 3Y Chg in Earnings to Price**

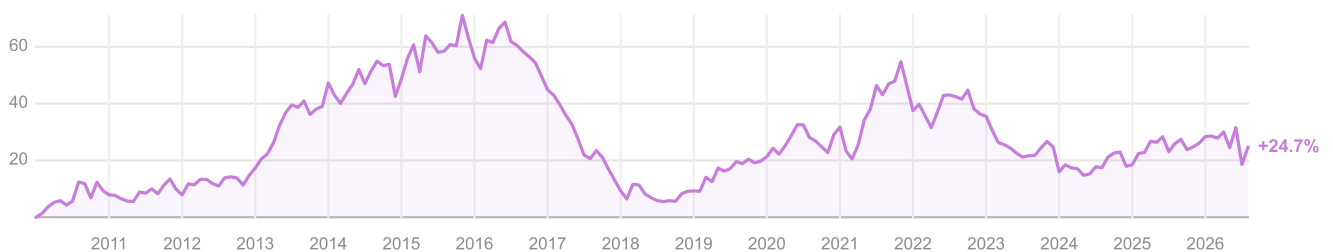
Historical Growth

CAGR +1.2%

**167 ● Log TTM Sales**

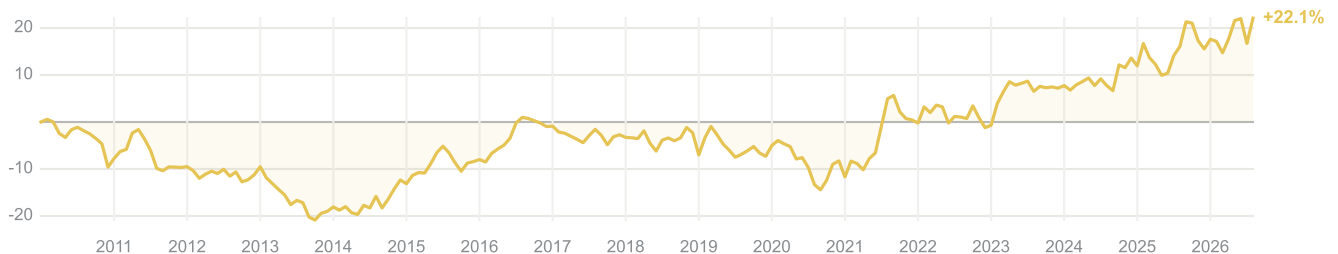
Size

CAGR +1.2%

**168 ● Degree of Financial Leverage**

Capital Efficiency

CAGR +1.2%



169 ● **3Y Chg in Sales to Price**

Historical Growth

CAGR +1.2%

170 ● **Asset Growth - 1 Year**

Capital Efficiency

CAGR +1.2%

171 ● **EBITDA to Price**

Valuation

CAGR +1.2%

172 ● **Volatility adjusted 12M return**

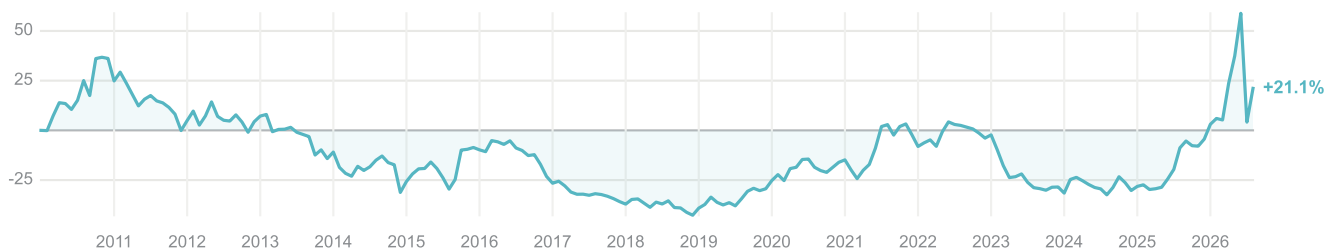
Price Momentum

CAGR +1.2%

173 ● **12M Realized Price Volatility**

Volatility

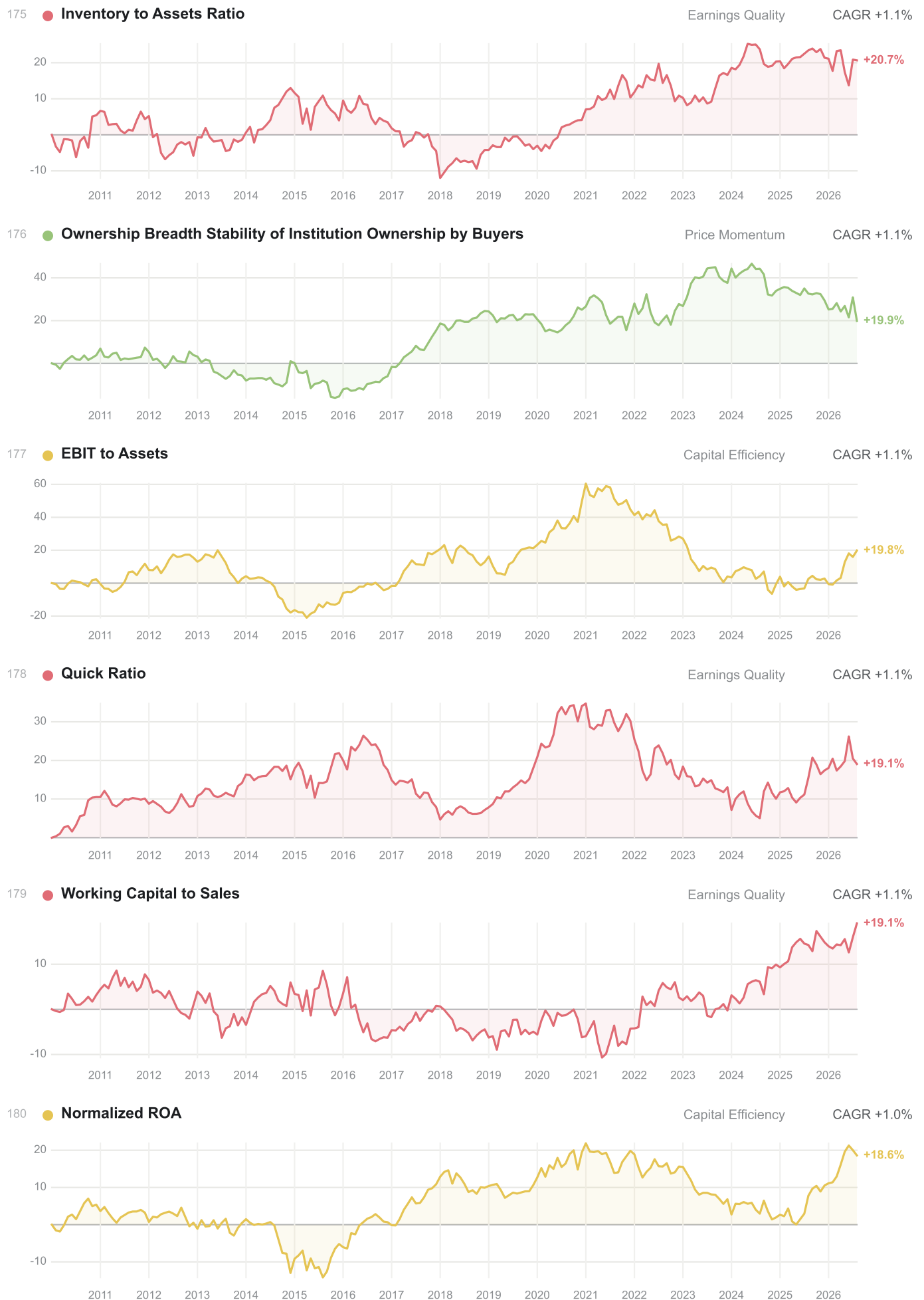
CAGR +1.2%

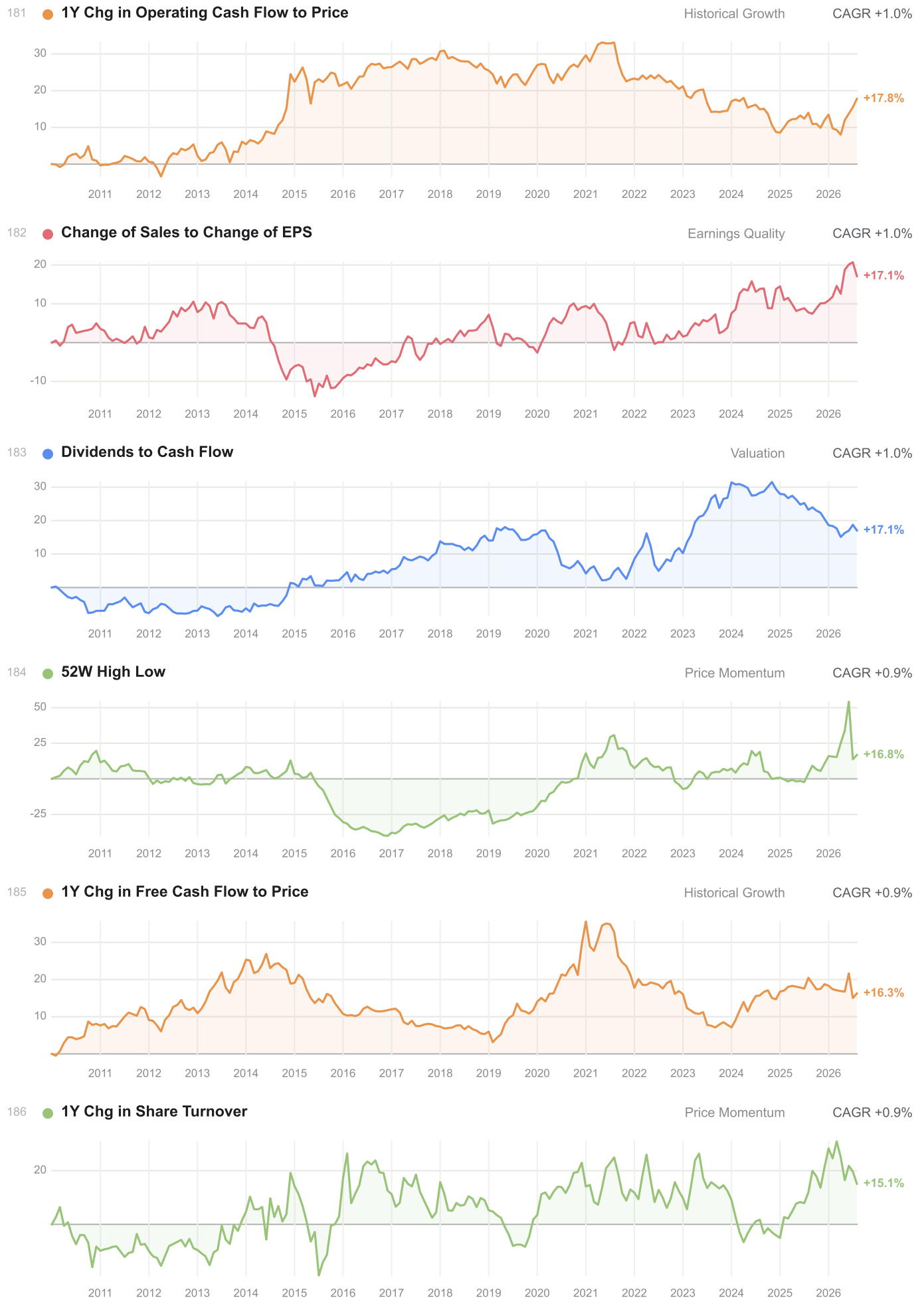
174 ● **ROA 60 Month Slope**

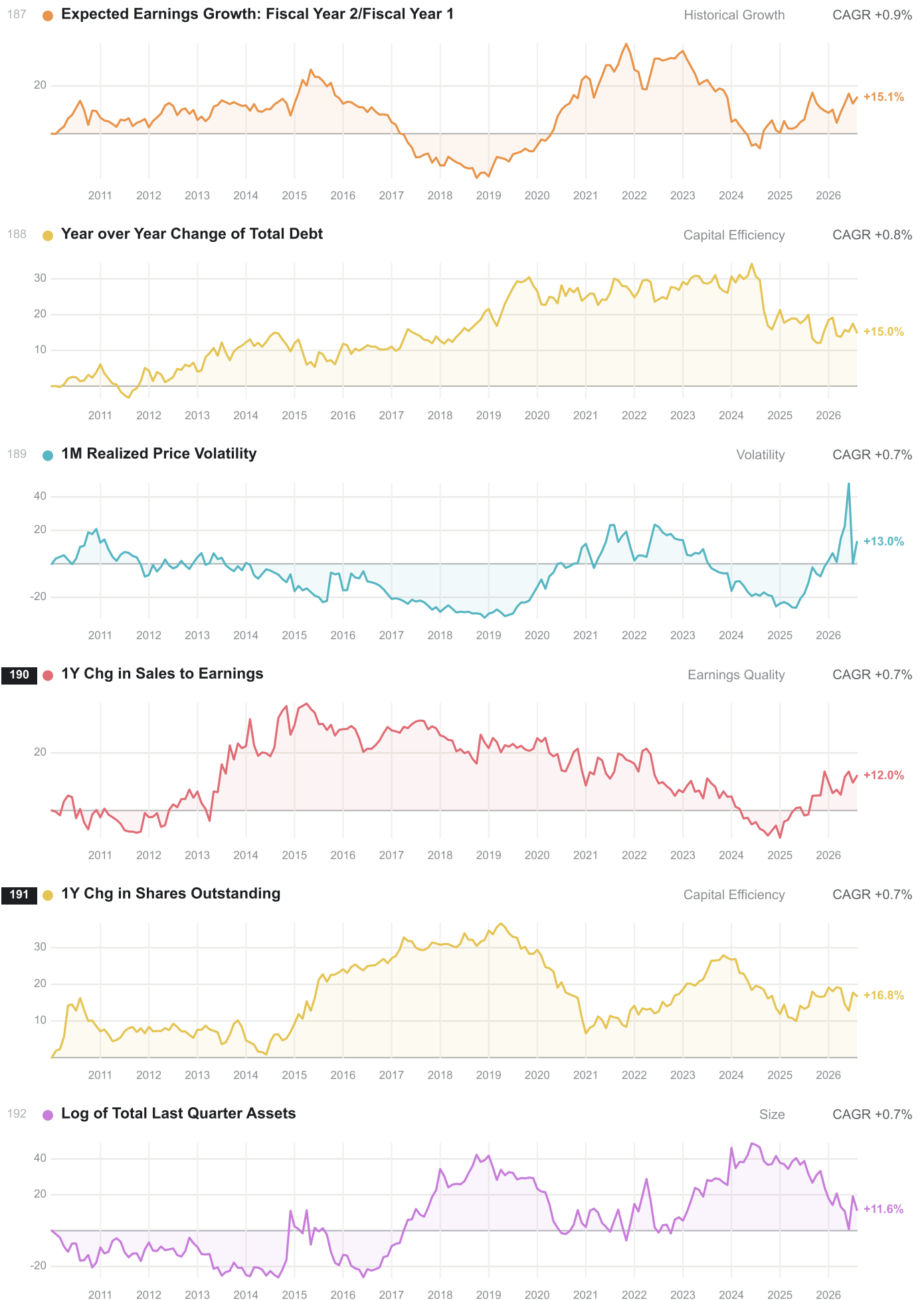
Earnings Quality

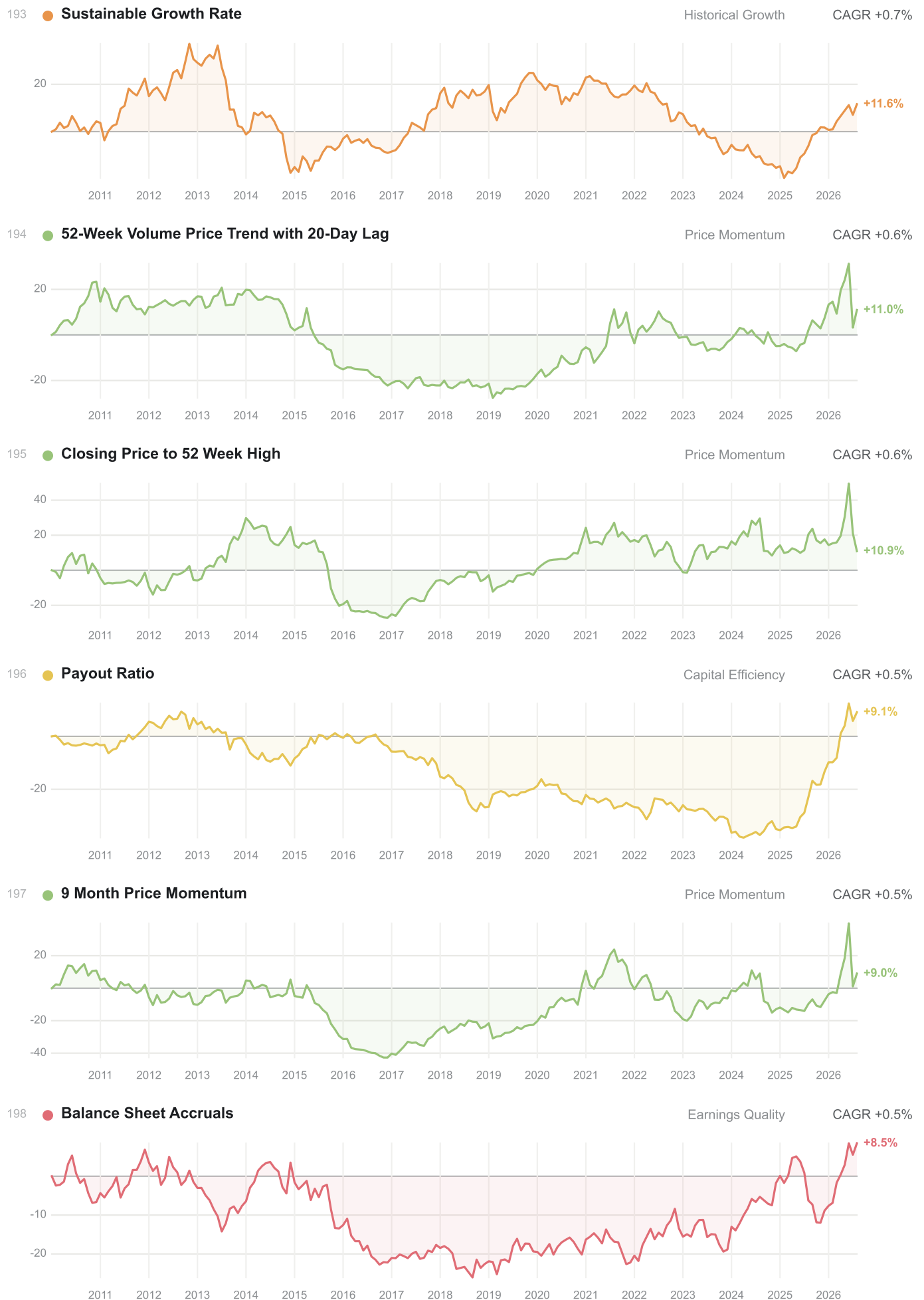
CAGR +1.1%

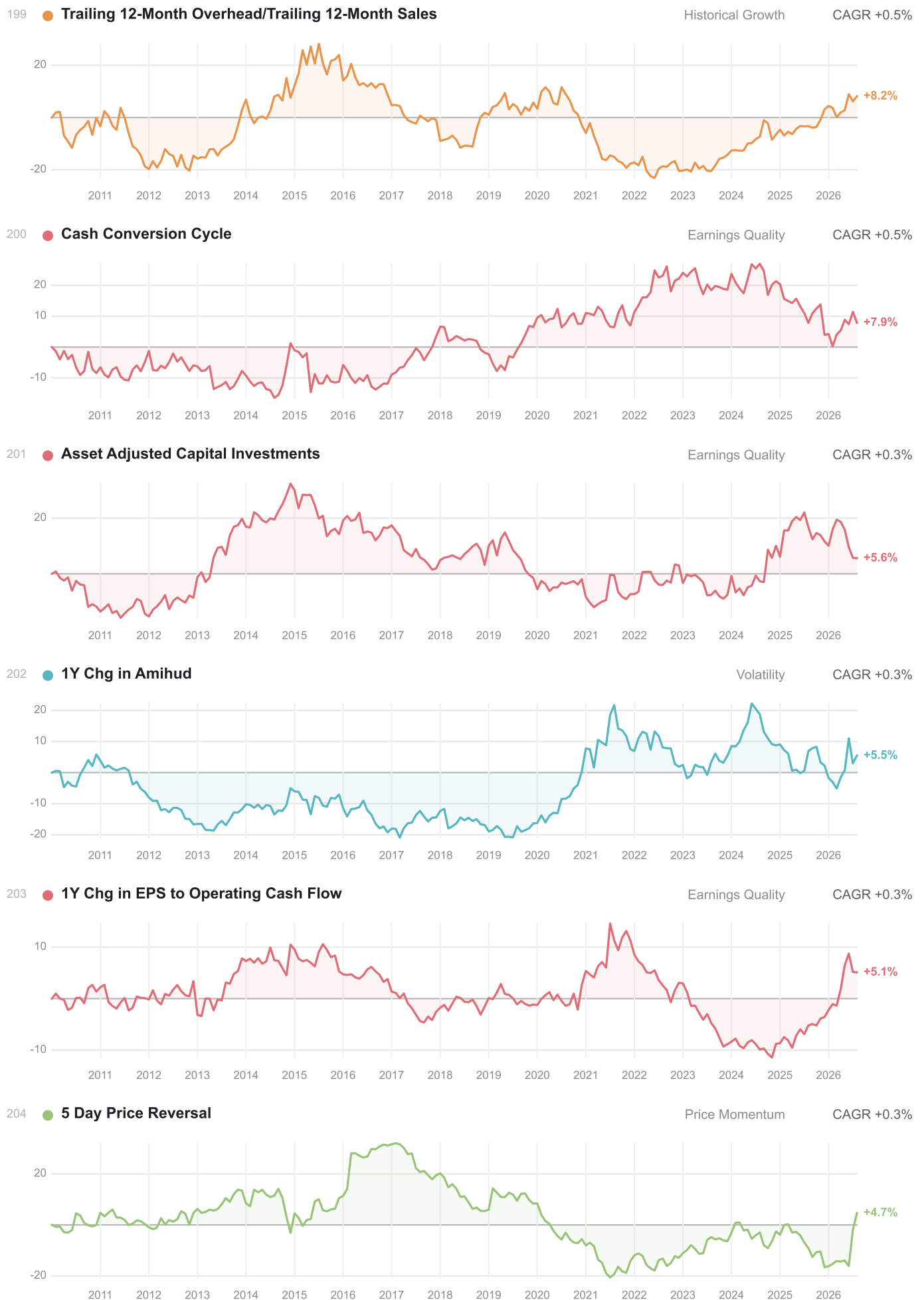












205 ● **6M Chg in 12M CAPM Alpha**

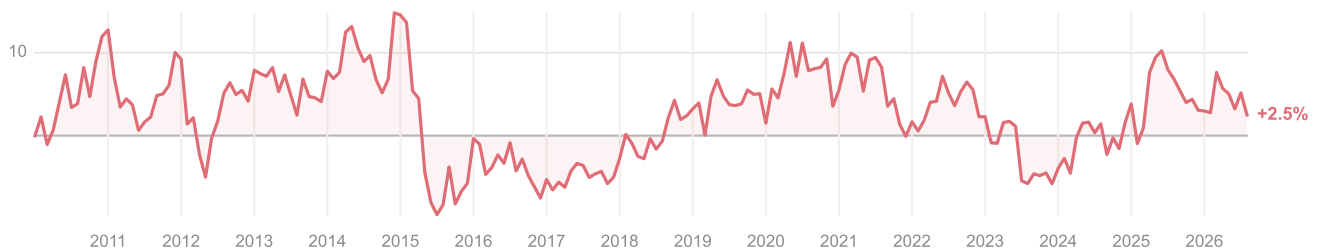
Price Momentum

CAGR +0.3%

206 ● **Inventory Turnover**

Earnings Quality

CAGR +0.1%

207 ● **Analyst Dispersion for FY2 EPS**

Analyst Expectations

CAGR -0.1%

208 ● **Change in Number of Holders**

Price Momentum

CAGR -0.2%

209 ● **1M Price Reversal**

Price Momentum

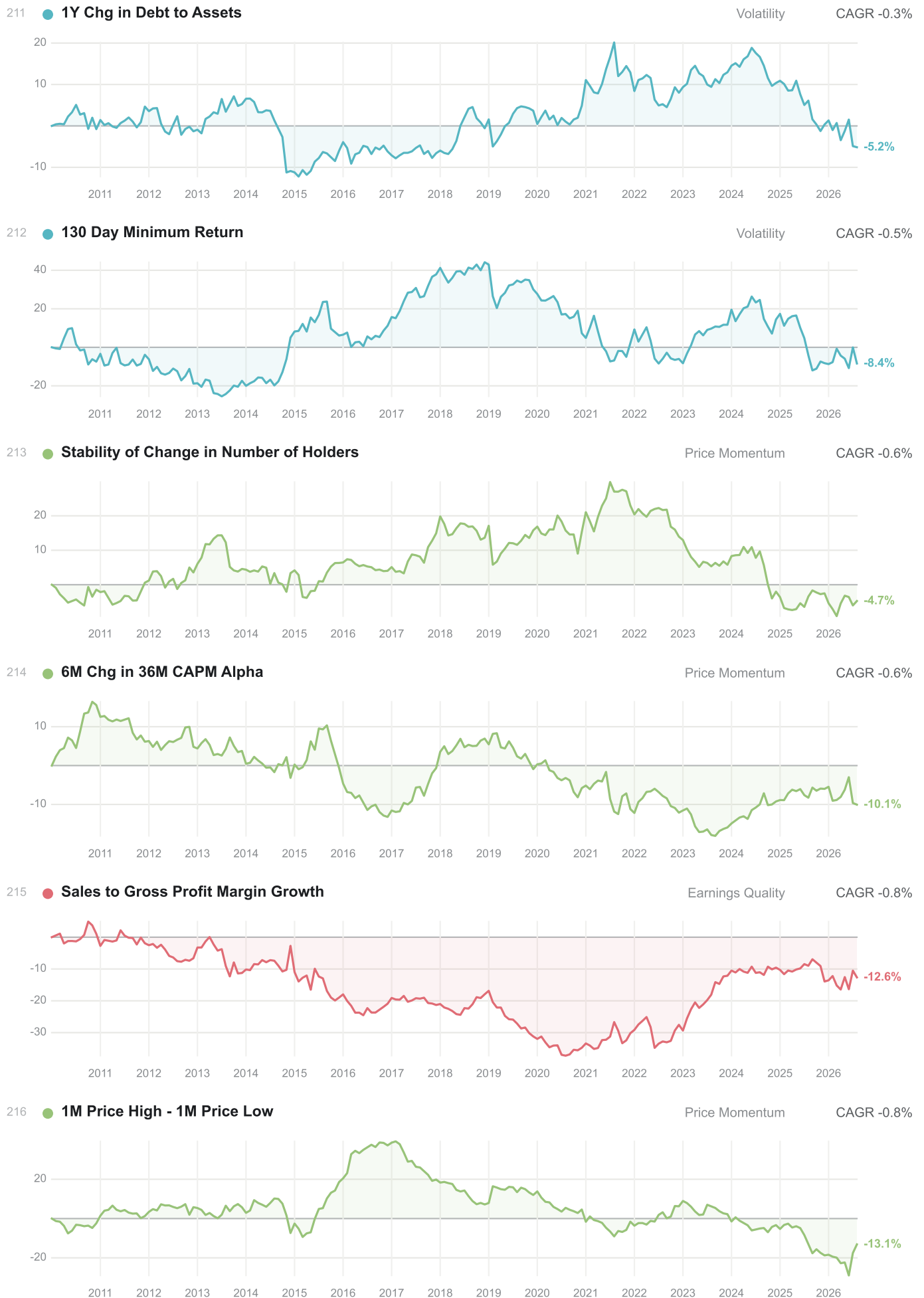
CAGR -0.2%

210 ● **4-Week Change in 12-Month Forward Earnings Consensus Estimate/Price**

Valuation

CAGR -0.3%





217 ● **26W Relative Price Strength**

Price Momentum

CAGR -0.9%

218 ● **20-Day Lane's Stochastic Indicator**

Price Momentum

CAGR -1.0%

219 ● **Revenue Estimate Dispersion**

Analyst Expectations

CAGR -1.0%

220 ● **14-Day Relative Strength Index**

Price Momentum

CAGR -1.2%

221 ● **6M Chg in 18M CAPM Alpha**

Price Momentum

CAGR -1.9%

222 ● **EBIT Estimate Dispersion**

Analyst Expectations

CAGR -2.1%

